# 0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

11 November 2026 · T. Santos, L. Batista

**Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group**

| Field | Value |
|---|---|
| Group | Group 1 · Fernão de Oliveira |
| Members (student number) | 106238 · 107304 · 106645 |
| Baseline | B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week |
| Specialization 1 | SP01 · Longitudinal Strength Calculation |
| Specialization 2 | SP02 · Propulsion Matching |
| Deadline | Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com |
| ZIP name | Group1_Project.zip |
| Presentation | Dec 18, 2026, 15 minutes per group |

## Deliverable Groups D1–D11

| # | Deliverable Group | Description |
|---|---|---|
| D1 | Design Basis | 130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked |
| D2 | Reference Ship Database | At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds |
| D3 | Ship Model | Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance |
| D4 | Voyage and Financial Model | Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input |
| D5 | Validation Note | Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed |
| D6 | Optimization Report | Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements |
| D7 | Compliance Sheet | EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules · stability while the crane lifts · freeboard · draught at every port |
| D8 | General Arrangement and Hull Form | Deck container layout and crane reach · battery slots aft · hull form of your choice, justified · A3 sketch (profile and midship section) · 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior · hydrostatics check |
| D9 | Specialization Documents (2) | One individual report per specialization, with its required studies and drawings (see Your specializations) |
| D10 | Datasheet and Presentation | One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes |
| D11 | Code Flowchart (mandatory) | A flowchart of your code that shows how the information flows: inputs, each calculation step, the iterations and the outputs · symbols of ISO 5807, the standard for data, program and system flowcharts |

## Deliverables

| ID | Title | Deliverable | Status |
|---|---|---|---|
| STU-D1.1 | Owner's requirements and port limits | D1 | |
| STU-D1.2 | Environmental targets | D1 | |
| STU-D2.1 | Reference ship database with sources | D2 | |
| STU-D2.2 | Regressions and bounds | D2 | |
| STU-D3.1 | Ship model | D3 | |
| STU-D3.2 | Closed weight balance | D3 | |
| STU-D4.1 | Voyage model | D4 | |
| STU-D4.2 | Financial model | D4 | |
| STU-D5.1 | Validation against reference ships | D5 | |
| STU-D6.1 | Optimization | D6 | |
| STU-D7.1 | EEDI and CII | D7 | |
| STU-D7.2 | Freeboard and intact stability | D7 | |
| STU-D7.3 | Draught at every port | D7 | |
| STU-D8.1 | Hydrostatics check | D8 | |
| DWG-D8.1 | Profile (A3) | D8 | |
| DWG-D8.2 | Midship section (A3) | D8 | |
| DWG-D8.3 | 3D hull with containers, bulkheads and deckhouse | D8 | |
| STU-D10.1 | Datasheet (template 0.4) | D10 | |
| STU-D10.2 | Presentation, 15 minutes | D10 | |
| STU-D11.1 | Code flowchart (ISO 5807) | D11 | |
| STU-B.1 | Capacity count on own arrangement | D8 | |
| DWG-B.1 | Container and battery layout with crane reach | D8 | |
| STU-SP01.1 | Lightship and deadweight distribution | D9 · SP01 | |
| STU-SP01.2 | Four critical loading conditions | D9 · SP01 | |
| STU-SP01.3 | Still-water and wave bending moments and shear forces | D9 · SP01 | |
| STU-SP01.4 | Allowable stresses, bending and shear | D9 · SP01 | |
| STU-SP01.5 | Preliminary scantlings of the midship cross-section | D9 · SP01 | |
| STU-SP01.6 | Section modulus, moment of inertia and neutral axis | D9 · SP01 | |
| STU-SP01.7 | Allowable still-water bending moment and shear force vs loading conditions | D9 · SP01 | |
| DWG-SP01.1 | Midship cross-section | D9 · SP01 | |
| STU-SP02.1 | Optimum B-series propeller | D9 · SP02 | |
| STU-SP02.2 | Engine matching | D9 · SP02 | |
| STU-SP02.3 | PTO/PTI or electric operating modes | D9 · SP02 | |
| DWG-SP02.1 | Stern arrangement | D9 · SP02 | |

## Your specializations

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

### SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

### SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

- STU-SP02.1 Optimum Wageningen B-series propeller for the service speed.
- STU-SP02.2 Engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.3 The two-shaft electric modes (Baseline B).
- DWG-SP02.1 Stern arrangement: propeller, shaft line and rudder.

The full text of every item is in the Statement of Work.